

Simulation Inputs

IMU

- Gyroscope
- Accelerometer

Odometry

- Wheel Odometry

Initial State

- Position/Velocity
- Attitude
- EKF State
- Covariance

B Trajectories
× N Steps

§4.2 Kernel Fusion

Full-Loop Fusion

Single-kernel
Recursion

State Residency

Register-Resident
States

Matrix Specialization

Padded Tiles
Analytic Inverse

§4.3 Component-Aware Mix Precision

Component-Aware

Component Precision
Decoupling

Compile Specialization

Precision Path
Constexpr

Constrained Selection

Error-bounded
Latency

§4.4 Multi-Trajectory Parallelism

Trajectory Batching

Independent
Trajectories

Block Mapping

One Block
Per Scan

Residency Scaling

Register-Limited
Occupancy